

ST BEES HEAD, United Kingdom

WMO: 032100

Lat: 54.5178N

Lon: 3.6147W

Elev: 123

StdP: 99.86

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.2	-1.0	-6.0	2.3	0.4	-4.7	2.6	0.6	19.5	4.7	17.9	4.9	3.7	30	0.766

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	21.7	17.0	19.8	16.2	18.2	15.4	17.8	20.5	16.9	18.9	16.0	17.6	4.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.1	18.5	16.0	11.5	17.5	15.3	11.0	16.8	50.6	20.6	47.7	19.0	45.3	17.6	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.4	12.9	DB	-3.6	25.0	1.5	2.2	-4.6	26.6	-5.5	27.9	-6.4	29.1	-7.4	30.7	(4)
(5)			WB	-4.2	19.9	1.3	1.1	-5.2	20.7	-5.9	21.3	-6.6	21.9	-7.6	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	4.5	4.4	5.5	7.6	10.4	12.8	14.7	14.8	13.1	10.5	7.4	5.2		
	DBStd	4.46	2.44	2.36	2.27	2.44	2.62	2.11	2.00	1.75	1.82	2.32	2.61	2.88		
	HDD10.0	825	172	156	139	78	23	2	0	0	1	22	84	149		
	HDD18.3	3310	430	389	397	321	245	166	116	112	157	243	329	407		
	CDD10.0	563	0	0	1	7	37	87	146	149	95	37	4	0		
	CDD18.3	7	0	0	0	0	1	1	3	2	0	0	0	0		
	CDH23.3	24	0	0	0	0	3	2	12	8	0	0	0	0		
	CDH26.7	2	0	0	0	0	0	0	1	1	0	0	0	0		
(14) Wind	WSAvg	6.4	7.4	7.2	6.4	5.9	5.6	5.4	5.2	5.6	6.2	7.1	7.3	7.2		
(15) Precipitation	PrecAvg	1166	111	81	86	67	71	73	82	100	104	137	124	125		
	PrecMax	1453	176	161	143	114	125	193	145	213	179	249	248	286		
	PrecMin	908	12	21	42	19	12	34	47	18	24	38	71	35		
	PrecStd	169	41	41	28	25	29	34	25	49	36	68	46	57		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	10.7	13.5	18.2	22.1	22.9	24.9	24.2	20.2	16.9	13.6	11.2		
		MCWB	9.4	8.0	9.6	12.2	16.3	17.4	18.5	18.6	16.5	14.2	12.4	10.6		
	2%	DB	9.0	9.0	10.8	14.9	19.0	19.8	21.8	20.6	18.0	15.2	12.5	10.4		
		MCWB	8.4	8.0	8.3	10.5	14.9	15.9	17.4	16.9	15.2	13.8	11.7	9.9		
	5%	DB	8.4	8.2	9.2	12.5	16.5	17.7	19.6	18.7	16.7	14.4	11.6	9.6		
		MCWB	7.8	7.4	7.8	9.5	13.0	14.9	16.4	16.0	14.7	13.3	10.9	9.1		
	10%	DB	7.7	7.5	8.3	11.0	14.5	16.2	17.9	17.4	15.8	13.6	10.9	8.9		
		MCWB	7.2	6.9	7.3	8.8	12.0	14.0	15.5	15.3	14.1	12.6	10.3	8.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	9.1	10.4	12.9	17.0	18.1	19.4	19.0	17.5	15.4	13.0	10.8		
		MCDB	9.7	9.5	11.9	17.5	20.3	20.9	23.3	22.3	19.0	16.0	13.1	10.9		
	2%	WB	8.7	8.3	9.0	11.1	15.2	16.4	17.9	17.5	16.1	14.3	12.0	10.2		
		MCDB	8.9	8.7	10.0	13.7	18.9	19.0	20.9	19.8	17.1	14.8	12.3	10.4		
	5%	WB	8.0	7.7	8.2	10.0	13.6	15.3	16.9	16.6	15.3	13.6	11.1	9.3		
		MCDB	8.3	8.1	9.0	11.9	15.9	17.4	19.2	18.1	16.2	14.1	11.5	9.6		
	10%	WB	7.3	7.0	7.6	9.2	12.4	14.2	15.9	15.8	14.6	12.9	10.4	8.5		
		MCDB	7.7	7.4	8.3	10.6	14.0	15.8	17.5	17.0	15.5	13.5	10.8	8.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.4	3.6	4.3	5.1	5.6	5.2	4.9	4.7	4.6	4.1	3.6	3.6		
		MCDBR	3.3	3.5	4.8	7.3	8.7	7.5	7.5	6.6	5.7	4.4	3.8	3.6		
	5% WB	MCWBR	3.6	3.2	3.8	5.0	5.9	5.1	5.0	4.4	4.1	4.0	4.1	4.0		
		MCWBR	3.4	3.1	4.2	6.0	7.6	6.8	6.7	5.5	4.8	4.0	3.7	3.6		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.335	0.360	0.378	0.382	0.378	0.379	0.367	0.356	0.342	0.319	0.310		
		taud	2.288	2.376	2.343	2.321	2.361	2.396	2.394	2.443	2.475	2.457	2.365	2.264		
	Ebn at Noon		607	751	811	847	863	870	860	852	813	735	617	536		
		Edh at Noon	59	75	96	112	114	111	109	98	84	69	54	49		
(44) All-Sky Solar Radiation	RadAvg		0.55	1.23	2.35	3.76	4.88	4.92	4.75	3.84	2.71	1.45	0.70	0.42		
	RadStd		0.07	0.14	0.21	0.36	0.38	0.48	0.48	0.28	0.23	0.17	0.07	0.04		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page